

OMU, Japan

WMO: 474050

Lat: 44.5806N

Lon: 142.9636E

Elev: 16

StdP: 101.13

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.2	-14.1	-18.8	0.7	-15.6	-17.0	0.8	-13.5	11.0	-3.2	9.6	-3.4	3.0	230	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.4	27.7	21.1	25.7	20.1	23.8	19.3	22.5	26.2	21.4	24.3	20.3	22.8	5.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.2	15.9	24.8	20.3	15.0	23.5	19.3	14.1	22.2	66.8	26.4	62.6	24.3	58.7	22.9	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.6	7.3	DB	-19.9	31.6	2.3	1.9	-21.5	33.0	-22.8	34.1	-24.1	35.2	-25.7	36.5	(4)
(5)			WB	-20.1	24.5	2.2	1.1	-21.7	25.3	-23.0	26.0	-24.3	26.6	-25.9	27.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-6.8	-6.9	-2.2	3.9	8.8	12.2	16.6	19.0	16.0	9.6	2.6	-3.8	(6)	
	DBStd	9.63	3.45	4.10	4.03	4.00	4.64	3.78	3.63	3.15	3.02	3.30	4.14	3.53	(7)	
	HDD10.0	2356	520	472	379	190	81	16	0	0	0	47	224	427	(8)	
	HDD18.3	4658	778	706	637	433	298	188	81	29	83	269	471	685	(9)	
	CDD10.0	832	0	0	0	7	43	83	203	278	181	36	2	0	(10)	
	CDD18.3	94	0	0	0	0	2	4	26	49	13	0	0	0	(11)	
	CDH23.3	501	0	0	0	1	26	40	145	237	52	0	0	0	(12)	
	CDH26.7	104	0	0	0	0	4	7	34	52	6	0	0	0	(13)	
(14) Wind	WSAvg	3.2	3.5	3.3	3.6	3.7	3.2	2.5	2.2	2.5	3.1	3.5	3.5	3.5	(14)	
(15) Precipitation	PrecAvg	955	50	40	43	50	64	68	118	129	143	89	88	74	(15)	
	PrecMax	1315	105	88	83	129	152	174	283	341	342	139	175	137	(16)	
	PrecMin	618	25	16	8	5	20	27	17	45	63	29	42	41	(17)	
	PrecStd	160	19	18	18	26	32	32	63	79	76	33	37	24	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.5	6.7	10.9	20.8	26.6	27.4	30.2	30.6	27.0	21.1	16.6	8.2	(19)	
		MCWB	1.1	3.0	5.9	11.5	15.3	18.3	23.2	23.4	20.1	15.0	10.9	5.2	(20)	
	2%	DB	1.1	3.6	7.7	16.5	22.6	23.8	27.1	28.0	24.7	18.8	13.5	5.3	(21)	
		MCWB	-0.5	0.7	3.8	9.4	13.6	16.5	21.8	21.9	19.1	13.7	9.2	3.1	(22)	
	5%	DB	-0.1	1.4	5.5	13.5	19.4	20.7	24.6	26.0	23.0	17.2	11.3	3.2	(23)	
		MCWB	-1.2	-0.9	2.3	7.7	12.5	15.5	20.5	21.3	18.4	12.8	7.8	1.4	(24)	
	10%	DB	-1.1	-0.4	3.6	11.0	16.5	18.2	22.1	24.0	21.4	15.7	9.4	1.7	(25)	
		MCWB	-2.4	-2.0	0.9	6.3	11.2	14.9	19.3	20.5	17.6	12.0	6.8	0.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	3.1	6.2	12.3	16.5	19.2	24.1	24.2	21.9	16.3	11.7	5.7	(27)	
		MCDB	3.0	6.0	10.3	18.7	23.6	25.0	28.6	29.4	25.3	19.4	15.7	7.5	(28)	
	2%	WB	0.0	0.8	4.1	9.8	14.5	17.5	22.4	22.9	20.5	14.7	9.6	3.4	(29)	
		MCDB	0.8	3.2	7.2	15.7	20.9	22.3	26.2	26.8	23.3	17.5	12.5	4.9	(30)	
	5%	WB	-1.2	-0.7	2.5	8.1	13.1	16.2	20.9	21.8	19.2	13.6	8.2	1.6	(31)	
		MCDB	-0.2	0.9	5.2	12.8	18.6	19.6	23.8	24.8	21.7	16.3	10.8	3.0	(32)	
	10%	WB	-2.3	-1.8	1.1	6.6	11.6	15.0	19.4	20.9	18.2	12.4	6.9	0.2	(33)	
		MCDB	-1.2	-0.5	3.3	10.6	15.5	17.8	21.8	23.4	20.6	15.1	9.3	1.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.8	7.7	8.4	8.5	7.0	5.9	6.4	8.6	9.1	7.3	7.0	(35)	
		MCDBR	7.9	9.9	10.0	13.3	14.4	12.2	10.3	10.1	10.4	11.0	10.7	8.2	(36)	
	5% WB	MCWBR	6.9	8.0	7.0	7.9	7.4	6.5	5.6	5.2	6.1	7.0	7.5	6.8	(37)	
		MCWBR	7.4	9.0	9.6	12.4	12.8	10.6	9.4	8.4	8.5	9.8	9.8	7.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.339	0.386	0.451	0.513	0.529	0.508	0.487	0.461	0.409	0.407	0.385	0.357	(40)	
		taud	2.028	1.847	1.783	1.805	1.868	2.018	2.146	2.239	2.329	2.173	2.126	2.042	(41)	
	Ebn at Noon	Ebn at Noon	729	754	759	752	756	775	787	795	810	740	675	657	(42)	
		Edh at Noon	107	155	190	203	196	170	148	130	110	112	98	97	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.25	2.14	3.32	4.40	4.83	4.85	4.53	4.19	3.56	2.46	1.36	1.01	(44)	
	RadStd	RadStd	0.13	0.30	0.30	0.37	0.39	0.51	0.57	0.28	0.30	0.14	0.14	0.09	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page